



Deep Blue beat Kasparov with mathematical efficiency, not intelligence

become a measure of humanity's intelligence – of our very humanity?

The reason why Deep Blue is such a milestone in programming history is not the fact that it defeated Kasparov, but how it defeated Kasparov. Chess, as with any game, is a game with a very clearly defined set of rules and win conditions, the kind of rules that are easy to program. The difficulty is in interpreting those rules to your advantage, reading your opponent, predicting his moves and responding to them.

Deep Blue could do none of that. It was a computer that was way more powerful than the average person had access to and worked by calculating all possible outcomes of any moves and picking the ones with the highest probability of success. There was no real “thought” behind what the computer did. It just calculated. Deep Blue wasn't designed to see any significance in an opponent's moves. All it ever did was accept the changed state of the chess pieces and then again calculate a move with the highest probability of success. And it could calculate thousands of these probabilities a second.

Mathematically, Garry Kasparov didn't stand a chance, but he still defeated Deep Blue 4-2 in 1996. How did that happen? As Kasparov explained, “In the end, that may have been my biggest advantage: I could figure out its priorities and adjust my play. It couldn't do the same to me. So although I think I did see some signs of intelligence, it's a weird kind, an inefficient, inflexible kind that makes me think I have a few years left.” This was in 1996.

In 1997, Kasparov lost to Deep Blue.

So what changed? A lot, apparently. Code was streamlined, new algorithms implemented, the computing platform enhanced, but none of these factors truly defeated Kasparov. The real nail in his coffin was a seemingly strange

What did IBM change? Did Deep Blue turn sentient? New evidence suggests that Gary Kasparov apparently thought as much. It's not clear exactly when the move occurred, but one specific move that Deep Blue made was supposedly so

“counter-intuitive” to Kasparov, that he was absolutely certain that he was dealing with a new form of intelligence. It was this move that completely threw off his game and lost him the rematch.

As it turns out, that “sentient” move was nothing more than a software bug. Deep Blue, confronted with innumerable possibilities, simply picked a move at random and defeated a former world champion.

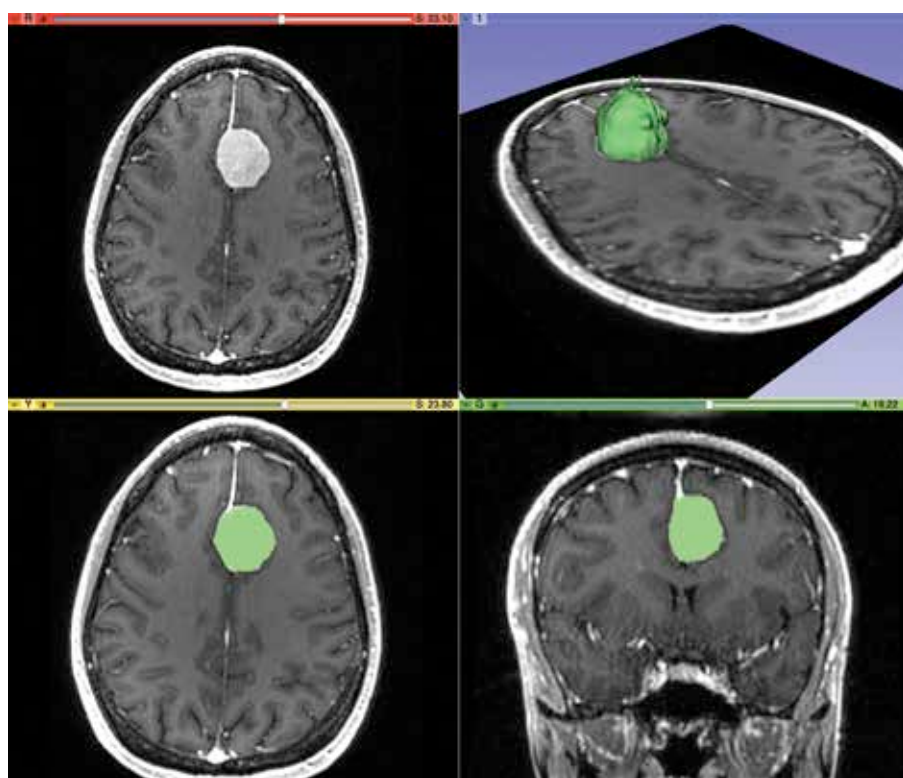
Fast forward to 2015 and new chess engines such as Stockfish are considered invincible, but the fact remains that the underlying principle is still the same. None of these engines are actually playing their opponent, they're simply following a logical pattern that will give them the best results.

This is not sentence. This is pure logical reasoning.

HEY, SIRI

Natural language input has long since been considered the holy grail of computing. The ability to actually talk to our computers and have them understand what we want is a magical feeling we would have considered impossible even twenty years ago.

We're very far from achieving anything resembling the future represented in



Our behaviour is shaped by Pavlovian conditioning in the pursuit of the dopamine rush